

Diffraction of Hermite–Gaussian beams from a sinusoidal conducting grating

Toshitaka Kojima

Department of Electro-Communication Engineering, Faculty of Engineering, Osaka Electro-Communication University, Neyagawa-shi, Osaka 572, Japan

Received June 16, 1989; accepted March 17, 1990

The diffraction of *E*-polarized Hermite–Gaussian beams of arbitrary order from a sinusoidal conducting grating is analyzed by using the plane-wave-spectrum representation for the incident beam field and the space harmonic solutions of the diffracted field for plane-wave incidence. It is shown that both the spot size and the location of the beam waist affect the diffraction characteristics. For the fundamental beam incidence, there exist major peaks in the far-field diffraction pattern, and the angle of the maximum value of each peak coincides with that of one of propagation space harmonics that will be produced when a plane wave is incident parallel to the beam axis. The decrease of the spot size makes the width of each peak broaden. For the first-order beam incidence, a deep dip is observed at each major peak. This dip is caused by the zero on the beam axis in the amplitude distribution of the incident beam field.

INTRODUCTION

Various analyses of the diffraction of electromagnetic or acoustic waves from periodic surfaces have been achieved since the first research by Rayleigh. Most of them have treated the case of plane-wave incidence, which is of course a basic and important subject.^{1–8} If the incident wave is a nonuniform one, such as a cylindrical wave, a spherical wave, or a Gaussian-beam wave, it will be required that the inhomogeneity of the amplitude and phase of the incident wave be taken into account. As far as I know, there seem to be many fewer papers^{9–15} that have treated the case for which the finite area of the periodical surface is illuminated by an incident wave such as a beam wave compared with those having treated the plane-wave incident case.

In a previous paper,¹⁵ I briefly reported the scattering of Gaussian beams from sinusoidal conducting surfaces. The purpose of this paper is to present the more detailed discussion of the diffraction of Hermite–Gaussian beams of arbitrary order from a sinusoidal conducting surface. A diffraction problem such as this is not only an interesting subject in the field of electromagnetic wave theory but also an important one relating to the surface measurement or diagnostics by microwave, millimeter-wave, laser, or ultrasonic beams, and especially to the development of various kinds of optical devices.

In order to utilize the plane-wave solution derived by DeSanto,⁴ we express the incident beam field as the superposition of various plane waves with different propagation constants; i.e., the plane-wave-spectrum representation is applied. The diffracted field is given by the integration with respect to the entire plane-wave solutions corresponding to all elementary plane waves that consist of the incident beam field. In order to examine how the diffraction characteristics are affected by the inhomogeneity of the incident wave, several numerical examples of the diffraction pattern are presented and discussed.

FORMULATION

The coordinate system of the problem is shown in Fig. 1. The diffraction surface is assumed to be a perfect conductor and its height profile $h(x)$ to be a sinusoid with period L and roughness amplitude d :

$$h(x) = -(d/2)[1 + \cos(2\pi x/L)]. \quad (1)$$

As an incident beam, we consider a two-dimensional Hermite–Gaussian beam of the order of m whose electric field is horizontally polarized. In a rectangular coordinate system (x', y', z') whose origin is located at $x = -x_0$ and $z = z_0$ and whose x' axis makes an angle θ_0 with the z axis, the field distribution of the incident beam is initially given by

$$E_{ym}^i(x' = 0, z') = H_m(\sqrt{2}z'/W_0)\exp(-z'^2/W_0^2), \quad m = 0, 1, 2, \dots \quad (2)$$

This is the two-dimensional version of the field distribution of a Hermite–Gaussian beam of the order of m at the beam waist $x' = 0$,¹⁶ where H_m denotes the Hermite polynomial of the order of m and the first three terms are given by $H_0(x) = 1$, $H_1(x) = 2x$, and $H_2(x) = 4x^2 - 2$ and W_0 denotes the beam radius or spot size. The time dependence $\exp(-i\omega t)$ is omitted throughout this discussion. The spectrum function of Eq. (2) is given by the Fourier transform of Eq. (2). Hence the incident beam field in the coordinate system (x', y', z') can be expressed by the inverse transform. By using the coordinate transformation between two coordinate systems (x', y', z') and (x, y, z) , we obtain the plane-wave-spectrum representation of the incident field in the coordinate system (x, y, z) as

$$E_{ym}^i(x, z) = \int_{-\infty}^{\infty} B_m(\alpha)\exp(ik\alpha x - ik\beta z)d\alpha, \quad (3)$$

where the spectrum function $B_m(\alpha)$ is given by

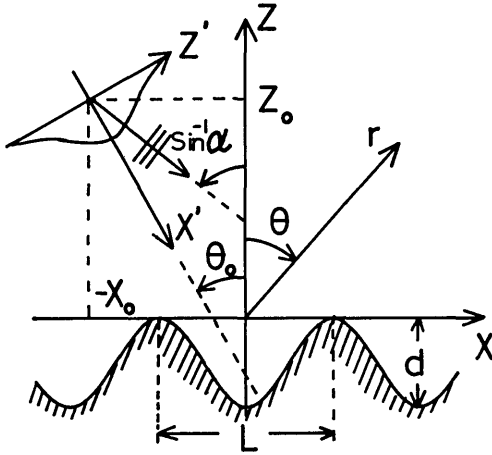


Fig. 1. Geometry of the problem.

$$B_m(\alpha) = \frac{kW_0}{2\sqrt{\pi}} (i)^m H_m \left[-\frac{kW_0}{2} (\alpha \cos \theta_0 - \beta \sin \theta_0) \right] \times \exp(ik\alpha x_0 + ik\beta z_0) \exp \left[-\frac{(kW_0)^2}{4} (\alpha \cos \theta_0 - \beta \sin \theta_0)^2 \right] \times \left(\cos \theta_0 + \frac{\alpha}{\beta} \sin \theta_0 \right), \quad (4)$$

and $\alpha^2 + \beta^2 = 1$. It should be noted that the term $B_m(\alpha) \exp(ik\alpha x - ik\beta z)$ in Eq. (3) represents an elementary plane wave with the angle of incidence $\theta_i = \arcsin \alpha$. For such an elementary plane-wave incidence, the diffracted waves are represented by the space harmonics

$$E_{ym}^s(x, z) = \sum_{n=-\infty}^{\infty} B_m(\alpha) R_n(\alpha) \exp(ik\alpha_n x + ik\beta_n z), \quad (5)$$

where $R_n(\alpha)$ is the diffraction coefficient of the n th-order space harmonic for the case for which a unit amplitude plane wave is incident on the sinusoidal surface. Considering the height profile in Eq. (1) together with the periodicity of the boundary value problem, we obtain the following grating equation:

$$\alpha_n = \alpha_0 + n\Lambda, \quad \beta_n = (1 - \alpha_n^2)^{1/2}, \quad n = \pm 1, \pm 2, \dots, \quad (6)$$

where $\Lambda = \lambda/L$, $I_m(\beta_n) \geq 0$, and λ is the wavelength. The diffraction coefficient R_n can be calculated by applying Green's theorem with the Dirichlet boundary condition on $h(x)$.^{4,13}

The diffracted wave for the beam-wave incidence can be obtained by integrating Eq. (5) with respect to α ; i.e.,

$$E_{ym}^s(x, z) = \sum_{n=-\infty}^{\infty} E_{ymn}(x, z), \quad (7)$$

where

$$E_{ymn}(x, z) = \int_{-\infty}^{\infty} B_m(\alpha) R_n(\alpha) \exp(ik\alpha_n x + ik\beta_n z) d\alpha. \quad (8)$$

If the observing point of the diffracted field is sufficiently far

from the origin, i.e., if $kr \gg 1$ and $r \gg W_0$, the integral of Eq. (8) can be evaluated by the saddle-point method.¹³ Then, in the polar coordinated system (r, θ) , the far-zone diffracted fields for all space harmonics are represented by

$$E_{ym}^s(r, \theta) = C(kr) \sum_{n=-\infty}^{\infty} \Psi_{mn}(\sin \theta - n\Lambda), \quad (9)$$

where

$$C(kr) = [2/(kr)]^{1/2} \exp(ikr - i\pi/4), \quad (10)$$

$$\Psi_{mn}(\sin \theta - n\Lambda) = B_m(\sin \theta - n\Lambda) R_n(\sin \theta - n\Lambda) \cos \theta, \quad (11)$$

and

$$R_n(\sin \theta - n\Lambda) = -\frac{\beta_0^{(n)}}{\beta_n^{(n)}} (i)^n \exp[i\tau_n^{+(n)}] \times \left\{ J_n[\tau_n^{+(n)}] + \sum_{l \neq 0} a_l J_{n-l}[\tau_n^{+(n)}] \right\}, \quad (12)$$

$$\beta_0^{(n)} = \{1 - [\alpha_0^{(n)}]^2\}^{1/2}, \quad \alpha_0^{(n)} = \sin \theta - n\Lambda, \quad \tau_n^{+(n)} = \Delta(\beta_0^{(n)} + \cos \theta), \quad \Delta = \pi d/\lambda. \quad (13)$$

a_l are the coefficients of Fourier expansion of the surface currents given by the solutions of the linear equations

$$J_\nu[\tau_\nu^{-(n)}] + \sum_{l \neq 0} a_l J_{\nu-l}[\tau_\nu^{-(n)}] = 0, \quad \nu = \pm 1, \pm 2, \dots, \quad (14)$$

where

$$\tau_\nu^{-(n)} = \Delta[\beta_0^{(n)} - \beta_\nu^{(n)}], \quad (15)$$

$$\beta_\nu^{(n)} = \{1 - [\alpha_\nu^{(n)}]^2\}^{1/2}, \quad \alpha_\nu^{(n)} = \alpha_0^{(n)} + \nu\Lambda, \quad (16)$$

and J_ν is the Bessel function of the order of ν .

The angular distribution of the scattered power is defined by the following function $G_m(\theta)$:

$$G_m(\theta) = \left| \sum_{n=-\infty}^{\infty} \Psi_{mn}(\sin \theta - n\Lambda) \right|^2. \quad (17)$$

NUMERICAL RESULTS AND DISCUSSION

In order to verify the accuracy of the computed values of $R_n(\alpha)$, I used the same criteria as those presented in Ref. 13.

Table 1. Errors of the Energy Conservation ϵ_M for Plane-Wave Incidence with $\theta_0 = 45^\circ$

L/λ	d/λ	ϵ_M for Matrix Size M			
		$M = 5$	$M = 10$	$M = 15$	$M = 20$
10.0	0.1	1.03×10^{-7}	1.35×10^{-7}	1.35×10^{-7}	1.35×10^{-7}
	0.5	2.92×10^{-7}	2.05×10^{-7}	2.64×10^{-7}	2.72×10^{-7}
	1.0	3.11×10^{-4}	2.62×10^{-4}	2.70×10^{-5}	2.69×10^{-5}
5.0	0.1	7.33×10^{-8}	8.92×10^{-8}	8.92×10^{-8}	8.92×10^{-8}
	0.5	1.60×10^{-5}	5.63×10^{-8}	8.31×10^{-8}	8.16×10^{-8}
	1.0	1.44×10^{-2}	2.62×10^{-4}	2.32×10^{-4}	2.30×10^{-4}

^a Criteria presented in Ref. 13.

Table 1 illustrates how the errors for the energy conservation ϵ_M depend on some selected values of period L , roughness amplitude d , and the size of matrices M at which the $\{a_n\}$ are computed after truncating the set of Eq. (14), where all parameters are normalized by the wavelength $\lambda = 2\pi/k$. It is seen from this table that it is possible to make the errors

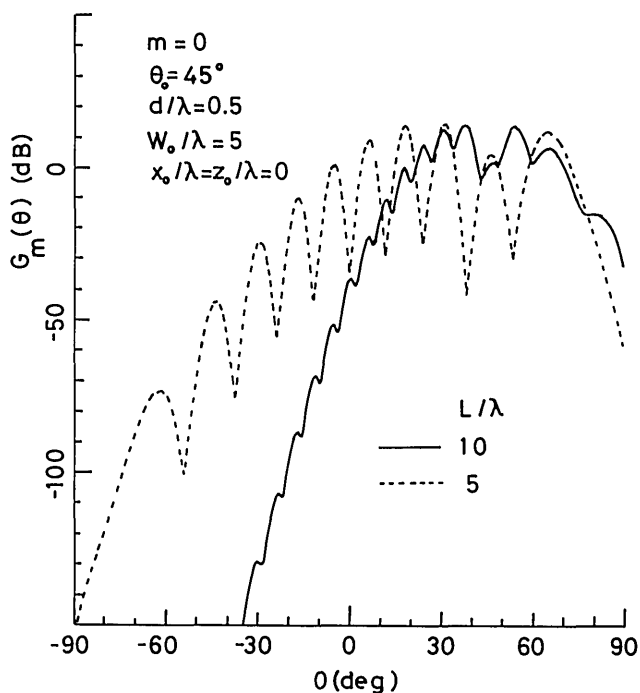


Fig. 2. Diffraction-pattern dependence on relatively large values of the period L for the obliquely incident fundamental beam ($m = 0$).

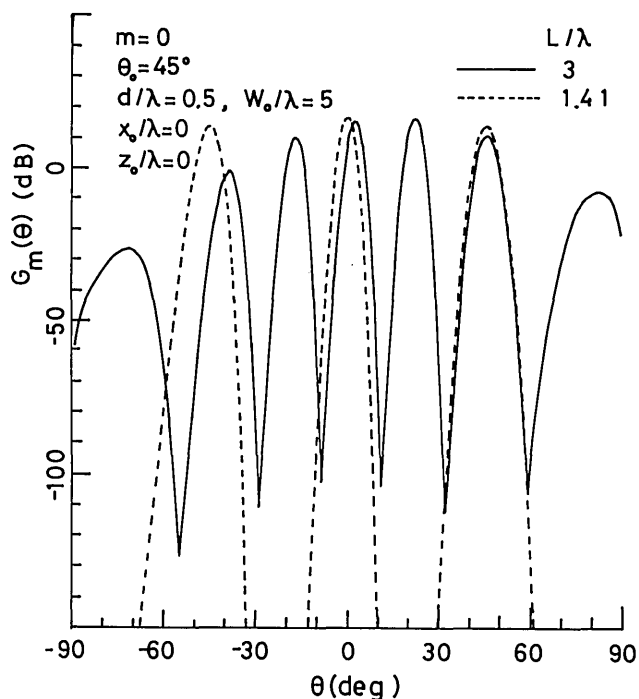


Fig. 3. Diffraction-pattern dependence on relatively small values of the period L for the obliquely incident fundamental beam ($m = 0$).

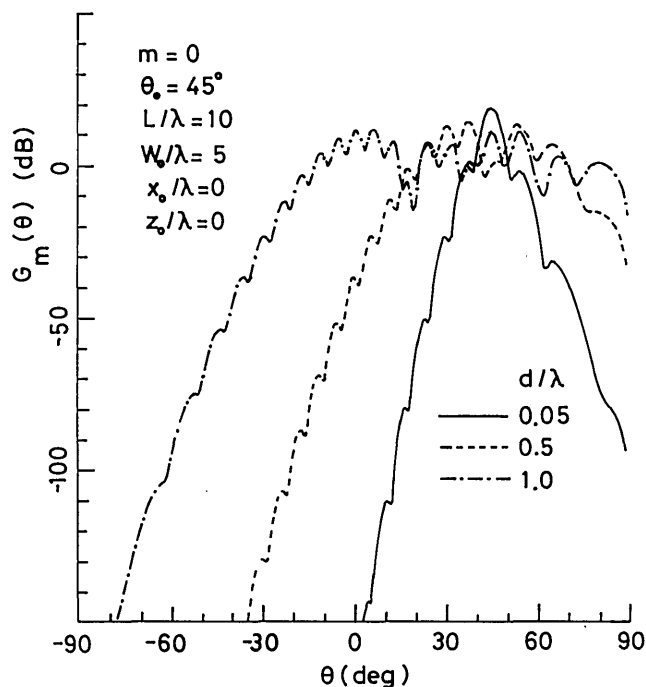


Fig. 4. Diffraction-pattern dependence on the roughness amplitude d for the obliquely incident fundamental beam ($m = 0$).

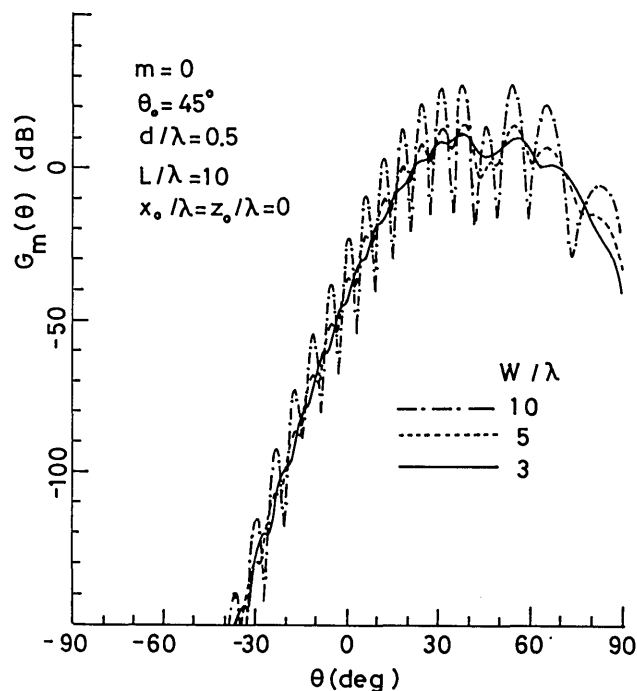


Fig. 5. Diffraction-pattern dependence on the spot size W_0 for the obliquely incident fundamental beam ($m = 0$).

sufficiently small by taking a large number of space harmonics into account. In the present calculations, the value of M needs to be 15 at least.

Several numerical examples of the diffraction pattern are shown and discussed. Figures 2 and 3 are examples of the diffraction pattern for the case in which the fundamental Gaussian beam is obliquely incident on the surface. They illustrate how the diffraction pattern depends on the period

L . We can see that the decrease of L causes the increase of nonspecular diffracted field components. In other words, the decrease of L is equivalent to the increase of the surface roughness.

Figure 4 is the diffraction pattern for several different values of the roughness amplitude d . This figure shows that the increase of d produces the increase of the nonspecular diffracted components, as was expected.

In order to examine how the spot size affects the diffraction characteristics, an example of the diffraction pattern for three different values of W_0 is shown in Fig. 5. It can be seen that the decrease of the spot size makes the height of peaks and the depth of dips in the diffraction pattern small and consequently the width of the peak becomes broad. This tendency is similar to the fact that the decrease of the number of corrugations results in the increase of the width of the diffracted beam.⁸ For the beam-wave incidence, even if the grating has an infinite extent, the illuminated area is not unbounded. Then the grating effect becomes small as the spot size is decreased. On the other hand, for the plane-wave incidence, if the extent of the grating is finite, the effect also becomes small as the number of corrugations of the finite grating is decreased. As a consequence, it can be said that this similar tendency comes from the finite area illumination.

Figure 6 shows how the diffraction pattern can be affected by the location of the beam waist. For the sake of illustration, we consider only the case for which the fundamental beam is normally incident on the surface. The dotted curve corresponds to the case for which the beam waist is located at $x = 0$ and $z = 0$; i.e., the incident beam illuminates the bottom of the grating symmetrically. It is obvious that the diffraction pattern is symmetric with respect to $\theta = 0$. On the other hand, the solid curve corresponds to the case for

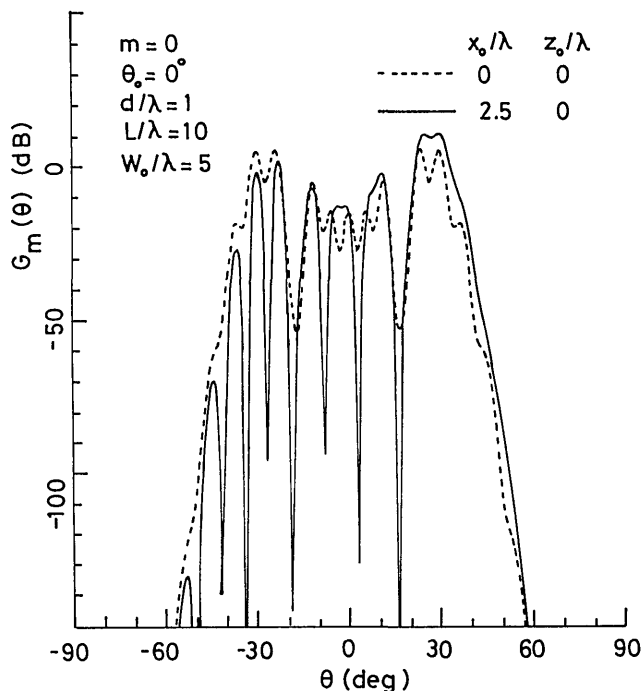


Fig. 6. Diffraction-pattern dependence on the displacement of the beam waist along the x axis for the normally incident fundamental beam ($m = 0$).

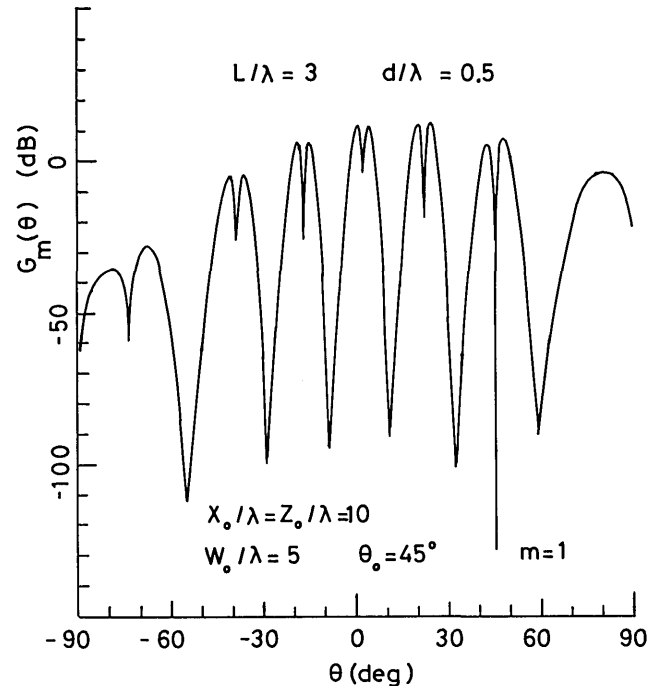


Fig. 7. An example of the diffraction pattern for the obliquely incident first-order beam.

which the beam waist is shifted along the x axis by 2.5λ from the point $x = 0$ and $z = 0$; i.e., the incident beam illuminates the bottom asymmetrically. In this figure a considerable change of the diffraction pattern can be observed. That is, not only does the symmetry of the diffraction pattern disappear, but also many peaks and dips appear.

Shown in Fig. 7 is an example of the diffraction pattern for the case for which the first-order Hermite-Gaussian beam is obliquely incident on the grating. We can observe a deep dip at a major peak located at the angle corresponding to each propagating space harmonic that will be produced if a plane-wave component propagating parallel to the beam axis is incident. It is obvious that this dip is caused by the zero on the beam axis in the amplitude distribution of the incident-beam field. This tendency is clearly apparent in the major peak at the specular angle. On the contrary, there is no dip in the major peak at approximately 80 deg. This peak does not correspond to any propagating space harmonic produced by the axial plane-wave component of the incident beam. However, we can confirm that this peak corresponds to the first-order propagating space harmonic for the elementary plane-wave component incident on the surface at an angle approximately of 40.7 deg.

CONCLUSION

The diffraction of two-dimensional Hermite-Gaussian beams of arbitrary order from a conducting sinusoidal grating has been analyzed by using the plane-wave-spectrum representation and DeSanto's solution for the plane-wave version. The far-zone diffracted field is obtained, and the numerical examples of the diffraction pattern are presented.

The results show that both the spot size and the location of the beam waist affect the diffraction characteristics. For the fundamental beam incidence, there exist major peaks in

the diffraction pattern and the angle of the maximum of each peak corresponds to one of propagating space harmonics that will be produced if a plane wave is incident parallel to the beam axis. The decrease of the spot size makes the width of each peak broaden. For the first-order beam incidence, a deep dip is observed at each major peak. This dip is caused by the zero on the beam axis in the amplitude distribution of the incident beam field. This tendency becomes clearly apparent in the major peak at the specular angle.

In the present analysis, the saddle-point method was used to evaluate Eq. (8), because we have been concerned with only the far-zone diffracted field. However, if one were interested in the field behavior in the intermediate and near regions, one would have to use the more complex evaluation techniques as given by Tamir and Bertoni.⁹ In such a case, one may also be able to predict some interesting and important behavior such as lateral displacement and distortions of the diffracted beam field. An analysis such as this is beyond the scope of the present paper and is left as a further problem to be studied.

REFERENCES

1. P. Beckmann and A. Spizzichino, *Scattering of Electromagnetic Waves from Rough Surfaces* (Pergamon, New York, 1963), Chap. 4.
2. A. Hessel and A. A. Oliner, "A new theory of Wood's anomalies on optical gratings," *Appl. Opt.* **4**, 1275-1297 (1965).
3. K. A. Zaki and A. R. Neureuther, "Scattering from a perfectly conducting surface with a sinusoidal height profile: TE polarization," *IEEE Trans. Antennas Propag.* **AP-19**, 208-214 (1971).
4. J. A. DeSanto, "Scattering from a sinusoid: derivation of linear equations for the field amplitudes," *J. Acoust. Soc. Am.* **57**, 1195-1197 (1975).
5. G. Whitman and F. Schwering, "Scattering by periodic metal surfaces with sinusoidal height profiles—a theoretical approach," *IEEE Trans. Antennas Propag.* **AP-25**, 869-876 (1977).
6. R. Petit, ed., *Electromagnetic Theory of Gratings* (Springer-Verlag, New York, 1980).
7. T. K. Gaylord and M. G. Moharam, "Analysis and applications of optical diffraction by gratings," *Proc. IEEE* **73**, 894-937 (1985).
8. E. V. Jull, D. C. W. Hui, and P. Facq, "Scattering by dual-blazed corrugated conducting strips and small reflection gratings," *J. Opt. Soc. Am. A* **2**, 1049-1056 (1985).
9. T. Tamir and H. L. Bertoni, "Lateral displacement of optical beams at multilayered and periodic structures," *J. Opt. Soc. Am.* **61**, 1397-1413 (1971).
10. R. S. Chu and T. Tamir, "Bragg diffraction of Gaussian beams by periodically modulated media," *J. Opt. Soc. Am.* **66**, 220-226 (1976).
11. R. S. Chu and T. Tamir, "Diffraction of Gaussian beams by periodically modulated media for incidence close to a Bragg angle," *J. Opt. Soc. Am.* **66**, 1438-1440 (1976).
12. R. S. Chu, J. A. Kong, and T. Tamir, "Diffraction of Gaussian beams by a periodically modulated layer," *J. Opt. Soc. Am.* **67**, 1555-1561 (1977).
13. A. K. Jordan and R. H. Lang, "Electromagnetic scattering patterns from sinusoidal surface," *Radio Sci.* **14**, 1077-1088 (1979).
14. M. G. Moharam, T. K. Gaylord, and R. Magnusson, "Bragg diffraction of finite beams by thick gratings," *J. Opt. Soc. Am.* **70**, 300-304 (1980).
15. T. Kojima and E. Okumura, "Scattering of Hermite-Gaussian beams from a sinusoidal surface," in *Proceedings of the International Symposium on Antennas and Propagation* (Institute of Electronics and Communications Engineers, Tokyo, 1985), Vol. 3, pp. 835-838.
16. D. Marcuse, *Light Transmission Optics* (Van Nostrand Reinhold, New York, 1972), Chap. 6.